Unit 6:Regression

Introduction

Correlation gives magnitude and direction of relationship between two variables. After knowing correlation one may be interested in estimating the value of one variable when value of other is known. E.g. we known that expenditure on advertisement and sale of commodity are correlated, using this relation we may find expected amount of sale for given expenditure on advertisement. Thus there is need to study a technique by which we can estimate value of one variable for known value of other variable , when both the variables are correlated. One such technique is regression analysis. **Definition- Regression-** The method of estimating value of one variable for known value of other variable and when both the variables are correlated.

In regression there are two types of variables, a variable whose value is to be estimated is called as dependent variable and a variable whose value is known is called independent variable.

Lines of regression-

The line of regression is the line which gives the best estimate of one variable for any known value of othervariable. It is obtained as follows.

After drawing the scatter diagram a straight line is drawn through all points in such a way that it is very close to each point. Such a line is called regression line or line of best fit. It is obtained by minimizing the distances of the points from the line. These distances are measured in two ways i) vertically ii) horizontally, therefore we have two lines of regression.

1)Line of regression of Y on X-

If we minimize the vertical distances (i.e. distances parallel to Y-axis) of the points from the line, we get a line which is called line of regression of Y on X.

In this case Y is dependent and X is independent variable. This line is used to estimate value of Y for known value of X, the equation of this line of regression is written as $Y = a + bX$.

2)Line of regression of X on Y-

If we minimize the horizontal distances (i.e. distances parallel to X-axis) of the points from the line, we get a line which is called line of regression of X on Y.

In this case X is dependent and Y is independent variable. This line is used to estimate value of X for known value of Y, the equation of this line of regression is written as $X = a + bY$.

Note – if the correlation is perfect (i.e. $r = \pm1$), then both the lines of regressions are coincides i.e. we get only one line of regression.

Derivation of line of line of regression by least square method-

Let (x_i, y_i), i = 1,2,3,.......n be the n pairs of observations on bivariate (X,Y)

And let the equation of line of regression Y on X is Y = a + bX ....(1)

To fit equation (1) to given data i.e. to obtain the values of constants 'a' and 'b' by using given data.

Note that y_i be the i[th] observed value of Y and value of Y obtained from equation (1) for known value of X = x_i, is denoted by $\hat{y}_i$, called as linear estimate of Y and given by $\hat{y}_i = a + b x_i$ ....(2)

The equation of line (1) is expected to be best or ideal fit, if $y_i - \hat{y}_i = 0$

Therefore we find constants 'a' and 'b' such that the sum of squares of deviations of observations y_i from their regression estimate $\hat{y}_i$ is least.i.e. $\sum(y_i - \hat{y}_i)^2$ is least (minimum), so this meathod is known as least square method.

Let $S = \sum(y_i - \hat{y}_i)^2 = \sum(y_i - a - b x_i)^2$ ....[by (2)

Note that the differential equations $\frac{dS}{da} = 0$ and $\frac{dS}{db} = 0$ gives minimum value of 'a' and 'b' if $\frac{d^2S}{da^2} > 0$ and $\frac{d^2S}{db^2} > 0$ ....(3)

Now $\frac{dS}{da} = -2\sum (y_i - a - b x_i)$ and $\frac{d^2S}{da^2} = -2\sum(-1) = 2n > 0$

$\frac{dS}{db} = -2\sum (y_i - a - b x_i)\cdot x_i$ and $\frac{d^2S}{db^2} = -2\sum(-x_i^2) = 2\sum x_i^2 > 0$

Thus conditions in (3) are satisfied

$\therefore \frac{dS}{da} = 0 \Rightarrow -2\sum (y_i - a - b x_i) = 0 \Rightarrow \sum y_i - na - b\sum x_i = 0$

$\Rightarrow \sum y_i = na + b\sum x_i$ ....(4)

and $\frac{dS}{db} = 0 \Rightarrow -2\sum (y_i - a - b x_i)\cdot x_i = 0 \Rightarrow \sum x_i y_i - a\sum x_i - b\sum x_i^2 = 0$

$\Rightarrow \sum x_i y_i = a\sum x_i + b\sum x_i^2$ ....(5)

Now, eq^n(4)$\Rightarrow a = \frac{\sum y_i}{n} - b\frac{\sum x_i}{n}$ $\Rightarrow a = \bar{y} + b\bar{x}$ ....(6)

put this in , eq^n(5), we get

$$\sum x_i y_i = (\bar{y} - b\bar{x})\sum x_i + b\sum x_i^2$$

$$\therefore \sum x_i y_i = (\bar{y} - b\bar{x})n\bar{x} + b\sum x_i^2 \text{.......................} [\because \sum x_i = n\bar{x}$$

$$\therefore \sum x_i y_i = n\bar{x}.\bar{y} - bn\bar{x}^2 + b\sum x_i^2$$

$$\therefore \sum x_i y_i - n\bar{x}.\bar{y} = b(\sum x_i^2 - n\bar{x}^2)$$

$$\therefore \frac{\sum x_i y_i}{n} - \bar{x}.\bar{y} = b\left(\frac{\sum x_i^2}{n} - \bar{x}^2\right)$$

$$\therefore \text{cov}(x, y) = b\sigma_x^2 \text{.......................} [\because \text{cov}(x, y) = \frac{\sum x_i y_i}{n} - \bar{x}.\bar{y} \quad and \quad \sigma_x^2 = \frac{\sum x_i^2}{n} - \bar{x}^2$$

$$\therefore b = \frac{\text{cov}(x, y)}{\sigma_x^2}$$

$$\therefore b = \frac{r.\sigma_y}{\sigma_x} \text{..............................} [\because \quad r = \frac{\text{cov}(x, y)}{\sigma_x \cdot \sigma_y}$$

Put this in eq^n(6), we get $a = \bar{y} - \frac{r.\sigma_y}{\sigma_x}.\bar{x}$

Now put values of a and b in eq^n(1), we get equation of line of regression of Y on X as

$$(y - \bar{y}) = \frac{r.\sigma_y}{\sigma_x}.(x - \bar{x})$$

Similarly the equation of line of regression Xof X on Y is

$$(x - \bar{x}) = \frac{r.\sigma_x}{\sigma_y}.(y - \bar{y})$$

Note –

1)The lines, of regression Y on X and X on Y are passing through the point $(\bar{x}, \bar{y})$, (i.e. the point of intersection)

2)the constant $\frac{r.\sigma_y}{\sigma_x}$ in the line of regression Y on X is called coefficient of regression Y on X and denoted by 'b_{yx}', thus $b_{yx} = \frac{r.\sigma_y}{\sigma_x} = \frac{cov(x,y)}{\sigma_x^2}$

Similarly the constant $\frac{r.\sigma_x}{\sigma_y}$ in the line of regression X on Y is called coefficient of regression X on Y and denoted by 'b_{xy}', thus $b_{xy} = \frac{r.\sigma_x}{\sigma_y} = \frac{cov(x,y)}{\sigma_y^2}$

Interpretation of regression coefficient –

The equation of line of regression Y on X is

$$(y-\bar{y}) = \frac{r.\sigma_y}{\sigma_x}.(x-\bar{x}) \text{ Where } \frac{r.\sigma_y}{\sigma_x} = b_{yx} \quad \Rightarrow y = a + b_{yx}x$$

This implies that for unit change in value of X, the value of Y will be changed by 'b_{yx} 'units. Thus b_{yx} is the rate of change of values of Y for unit change in X.

If byx is +ve, the increase in value of X is associated with increasein value of Y i.e. correlation between X & Y is +ve. On the other handIf byx is -ve, the increase in value of X is associated with decrease in value of Y i.e. correlation between X & Y is -ve.

Thus coefficient of regression gives the rate of change in the value of dependent variable per unit change in independent variable and algebraic sign of coefficient of regression determines whether the correlation is +ve or –ve.

Properties of regression coefficients-

1)Regression coefficients and correlation coefficient have same algebraic sign.

→The reg. coeff. Y on X is $b_{yx}= r.\sigma_x/\sigma_y$ and reg. coeff. Xon Y is $b_{xy}= r.\sigma_x/\sigma_y$Note that the ratios σ_x/σ_y and σ_x/σ_y are always +ve, therefore the sign of b_{yx} and $b_{xy}$is dependence upon the sign of 'r' only

$\therefore$ *if* r *is is* $+ve$ *then* $b_{yx} \& b_{xy}$ *are* $+ve$ *and if* r *is* $-ve$ *then* $b_{xy} \& b_{xy}$ *are* $-ve$

hence proof

2)Product of regression coefficients is equal to square of correlation coefficient. OR The square root of product of regression coefficients is equal to correlation coefficient with sign of regression coefficients OR Geometric mean of regression coefficients is equal to correlation coefficient with sign of regression coefficients

→The reg. coeff. Y on X is $b_{yx} = r.\sigma_y/\sigma_x$ and reg. coeff. X on Y is $b_{xy} = r.\sigma_x/\sigma_y$

$$\therefore \quad b_{yx} \cdot b_{xy} = r.\frac{\sigma_y}{\sigma_x} \times r.\frac{\sigma_x}{\sigma_y} = r^2$$

$$\Rightarrow \quad \sqrt{b_{yx} \cdot b_{xy}} = r \quad (with \quad sign \quad of \quad regression \quad coefficients)$$

Hence proof

3)Both regression coefficients are not greater than 1 simultaneously OR If one of the regression coefficient is greater than one then other is less than one.

→The reg. coeff. Y on X is $b_{yx} = r.\sigma_y/\sigma_x$ and reg. coeff. X on Y is $b_{xy} = r.\sigma_x/\sigma_y$

$$\therefore \quad b_{yx} \cdot b_{xy} = r.\frac{\sigma_y}{\sigma_x} \times r.\frac{\sigma_x}{\sigma_y} = r^2$$

But we know that $-1 \leq r \leq +1$

$$\therefore \quad r^2 \leq 1 \quad \Rightarrow \quad b_{yx} \cdot b_{xy} \leq 1 \quad \Rightarrow b_{yx} \leq \frac{1}{b_{xy}}$$

$$\Rightarrow \quad if \quad b_{xy} > 1 \quad then \quad \frac{1}{b_{xy}} < 1 \quad \Rightarrow b_{yx} < 1$$

Hence proof

Note Both regression coefficients are less than 1 simultaneously

4)The A.M. of regression coefficients is greater than or equal to correlation coefficient.

$$Let us \quad assume \quad that \quad \frac{b_{yx} + b_{xy}}{2} \geq r$$

$$\Rightarrow b_{yx} + b_{xy} \geq 2r \quad \Rightarrow r\frac{\sigma_y}{\sigma_x} + r\frac{\sigma_x}{\sigma_y} \geq 2r \quad \Rightarrow \frac{\sigma_x^2 + \sigma_y^2}{\sigma_x \cdot \sigma_y} \geq 2 \quad \Rightarrow \sigma_x^2 + \sigma_y^2 - 2\sigma_x \cdot \sigma_y \geq 0$$

$$\Rightarrow (\sigma_x - \sigma_y)^2 \geq 0$$

Which is trivially true, hence above assumption is true.

5)Effect of change of origin and scale on regression coefficient.

→Let (x_i, y_i), i = 1,2,3,.......n be the n pairs of observations on bivariate (X,Y)

$\therefore$ *The regression coefficient Y on X is* $b_{YX} = r\dfrac{\sigma_Y}{\sigma_X}$

and regression coefficient X on Y is $b_{XY} = r\dfrac{\sigma_X}{\sigma_Y}$

Let us define $u_i = \dfrac{x_i - A}{B}$ *and* $v_i = \dfrac{y_i - C}{D}$ *where* A, B, C, D *are constants*

Now, we know that $r_{xy} = r_{uv}$, $\sigma_y = D.\sigma_v$, $\sigma_x = B.\sigma_u$

$$\therefore b_{YX} = r_{uv}.\frac{D.\sigma_v}{B.\sigma_u} = \frac{D}{B}b_{vu} \quad and \quad b_{XY} = \frac{B}{D}b_{uv}$$

$\Rightarrow$ *Regression coefficient is independent of change of origin but depends on change of scale*

Non-linear Regression-

Correlation gives the degree of linear relationship between two variables. Regression uses this linear relationship and estimate the value of one variable for known value of other variable. Note that reliability of regression estimate is depends upon the value of 'r' i.e. if correlation is perfect or high degree then regression estimate is more reliable.

Consider the following example-

X :	-2	-1	0	1	2
Y :	4	1	0	1	4

$\Rightarrow \sum X = 0, \sum Y = 10, \sum XY = 0$

$\Rightarrow \text{cov}(X,Y) = \dfrac{\sum XY}{n} - \overline{X}.\overline{Y} = 0$

$\Rightarrow r = 0$

$\Rightarrow$ *X and Y are uncorrelated, but we can clearly observed that* $Y = X^2$

Thus when variables are uncorrelated it only mean that there is no linear relationship between the two variables, therefore in case where correlation is zero or low degree the linear regression is will be of no use for estimation of dependent variable.

In this case relation involving second degree or higher degree, exponential, logarithmic etc. may gives satisfactory regression estimate. The regression equation other than linear equation (i.e. second degree or higher degree, exponential, logarithmic) is called non-linear regression. Non-linear regression represents the curve, so it is called regression curve.

@ Fitting of second degree curve-

Let (x_i, y_i), i = 1,2,3,.......n be the n pairs of observations on bivariate (X,Y) and X is independent while Y is dependent variable

let the general form of second degree curve will be $Y = a + bX + cX^2$ ....(1)

To fit equation (1) to given data i.e. to obtain the values of constants 'a' , 'b' and 'c' by using given data. This is done by least square principal

Note that y_i be the i^{th} observed value of Y and value of Y obtained from equation (1) for known value of $X = x_i$, is denoted by $\hat{y}_i$, called as linear estimate of Y and given by

$\hat{y}_i = a + b x_i + c x_i^2$.................(2)

The equation (1) is expected to be best or ideal fit, if $y_i - \hat{y}_i = 0$

Therefore . $\sum(y_i - \hat{y}_i)^2$ isthe error in estimation

Let $S = \sum(y_i - \hat{y}_i)^2 = \sum(y_i - a - bx_i - cx_i^2)^2$ ....[by (2)

We have to estimate the values of constants a, b and c in such a way that S will be minimum, this can be done by solving the equations $\frac{dS}{da} = 0, \frac{dS}{db} = 0$ and $\frac{dS}{dc} = 0$

$$\therefore \frac{dS}{da} = 0 \Rightarrow 2\sum (y_i - a - bx_i - cx_i^2).(-1) = 0$$

$$\Rightarrow \sum y_i = na + b\sum x_i + c\sum x_i^2 \quad \text{.......................................(3)}$$

$$\frac{dS}{db} = 0 \Rightarrow 2\sum (y_i - a - bx_i - cx_i^2).(-x_i) = 0$$

$$\Rightarrow \sum x_i y_i = a\sum x_i + b\sum x_i^2 + c\sum x_i^3 \quad \text{.......................................(4)}$$

$$\frac{dS}{dc} = 0 \Rightarrow 2\sum (y_i - a - bx_i - cx_i^2).(-x_i^2) = 0$$

$$\Rightarrow \sum x_i^2 y_i = a\sum x_i^2 + b\sum x_i^3 + c\sum x_i^4 \quad \text{.......................................(5)}$$

Equations (3), (4) &(5) are called normal equations, solving them simultaneously, we get the values of constant a, s and c for which S will be minimum.

@ Fitting of exponential curve-

Let (x_i, y_i), i = 1,2,3,.......n be the n pairs of observations on bivariate (X,Y) and X is independent while Y is dependent variable

let the general form of exponential curve will be $Y = a + b^X$ ....(1)

To fit equation (1) to given data i.e. to obtain the values of constants 'a' and 'b' by using given data. This is done by least square principal

To apply least square principal, first make transformation by taking log of both sides of eq^n(1)

$\Rightarrow \log Y = \log a + X \log b$

put log Y = U, log a = A and log b = B

$\Rightarrow U = A + BX$..(2)

Thus the eq^n(2) looks like the eq^n of regression line U on X

Note that u_i be the i^{th} observed value of U and value of U obtained from equation (2) for known value of X = x_i, is denoted by $\hat{u}_i$, called as linear estimate of U and given by

$\hat{u}_i = A + Bx_i$..................(3)

Now, The equation (2) is expected to be best or ideal fit, if $u_i - \hat{u}_i = 0$

Therefore . $\sum(u_i - \hat{u}_i)^2$ is the error in estimation

Let $S = \sum(u_i - \hat{u}_i)^2 = \sum(u_i - A - Bx_i)^2$ ....[by (3)

We have to estimate the values of constants A and B in such a way that S will be minimum, this can be done by solving the equations $\frac{dS}{dA} = 0$ and $\frac{dS}{dB} = 0$

$$\therefore \frac{dS}{dA} = 0 \Rightarrow 2\sum (u_i - A - Bx_i).(-1) = 0$$

$$\Rightarrow \sum u_i = nA + B\sum x_i \quad \text{...(4)}$$

$$\frac{dS}{dB} = 0 \Rightarrow 2\sum (u_i - A - bx_i).(-x_i) = 0$$

$$\Rightarrow \sum x_i u_i = A\sum x_i + B\sum x_i^2 \quad \text{...(5)}$$

Equations (4) &(5) are called normal equations, solving them simultaneously, we get the values of constant A and B for which S will be minimum

Hence a = Antilog A and b = Antilog B

Ex1. Given: $\bar{X} = 53$, $\bar{Y} = 28$, $b_{yx} = -1.5$, $b_{xy} = -0.2$. Find i)r_{xy} ii)estimate y when x = 60 iii)estimate x when y = 30.

Ex2. The regression equations are 3X –Y -5 = 0 and 4X – 3Y = 0. Find i)A.M. of X & Y ii) r_{xy} iii) C.V. of X & Y if $\sigma_x = 2$.

Ex3. The regression equations are 8X – 10Y+66 = 0 and 40X – 18Y - 214 = 0. Find i)A.M. of X & Y ii) r_{xy} iii) C.V. of X & Y if $\sigma_x = 3$.

Ex4. The regression equations of X on Y is 3Y – 5X + 15 =0. If the mean of Y is 45 and the ratio of s.d. of X & Y is 3:4. Find mean of X and corr(X,Y).

Ex5. The regression equations are 3x + 2Y -26 = 0 and 6X + Y -31 = 0. Find i)A.M. of X & Y ii) estimate Y for X = 2.

Ex6. Given: X = 4Y + 8 and Y = kX +4 are lines of regression X on Y and Y on X respectively. Which of the following value of k is consistent i) k=1 ii) k= -1/8 iii) k= ½ iv) k= 1/8.

Ex8. If two lines of regression are 18X+Y=λ and 2X+Y=μ. The A.M. of X & Y are -1 & 2 respectively then find λ, μ and correlation coefficient between X & Y.

Ex9. Obtain equations of lines of regression and hence estimate Y when X= 15, estimate X when Y = 25

X:	4	5	6	7	8
Y:	2	3	4	9	12

Ex10. Given n= 10,ΣX=120. ΣY=220, ΣX^2=2288, ΣY^2= 5506, ΣXY=3467. Obtain regression $eq^{ns}$and estimate Y when X=16

Ex11. Given: n=10, ΣX=628. ΣY=550, ΣX^2=40376, ΣXY=33969. Obtain regression eq^n Y on X and estimate Y when X=25.

Ex12. Fit second degree parabola to the data given below and estimate Y for X= 1988

X:	1980	1981	1982	1983	1984
Y:	24	37	68	88	105

Ex13. Fit exponential equation $Y=ab^X$ to the following data and estimate Y when X=15

X:	2	4	6	8	10
Y:	10	20	40	80	200